Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms. The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U, f)], [I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 ([8, Corollary 10.3, Proposition 10.10(1)])**.** *Ψ and $\mu\Psi$ are both isomorphisms of graded semirings.*

Formalization review T001: generators of the unmeasured kernel. Keep the original ground $\mathbb{S}$ fixed. Put

$$T = \mathbf{K}\,\mathrm{RES}[*] \otimes_{\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}\,\Gamma[*].$$

The following is a mathematical presentation of $\ker\Psi$, rather than an assertion that its complete proof is already formalized in Lean. It concerns the unmeasured map only.

Take any $X \in \mathrm{RES}[r]$, any definable $\delta \in \Gamma$, and any definable function $c : X \longrightarrow \delta^\sharp$. No point of $\delta^\sharp$ is required to be definable over $\mathbb{S}$ alone. For any $U \in \Gamma[m]$, $m \geq 1$, and coordinate vector e_i, put

$$g_{X,c,U,i} = [X]_r \otimes \big([U]_m - [U + \delta e_i]_m\big).$$

Then the kernel has the homogeneous generating presentation

$$\boxed{\ker\Psi = \big(g_{X,c,U,i} : X, c, U, i \text{ as above}\big) \subseteq T.}$$

This is a potentially redundant family; no finite or minimal generating set is claimed. Equivalently, for each X use every δ for which $\delta^\sharp$ has a definable point *depending on* $x \in X$. One can make c a coordinate projection by replacing X with its graph, without changing its RES isomorphism class or its degree.

The generating bijections. Each generator maps to zero because the original-ground definable map

$$(x, u_1, \ldots, u_m) \longmapsto (x, u_1, \ldots, c(x)u_i, \ldots, u_m)$$

is an RV bijection from $X \times U^\sharp$ to $X \times (U + \delta e_i)^\sharp$. Its inverse divides by $c(x)$. The finite labels need not be preserved by an unmeasured RV morphism. If δ already has an RV representative definable over the original ground, the displayed difference is already zero in T.

Why these generators suffice. Normalize Gamma factors by integral unimodular changes of coordinates, and transfer their fixed Gamma coordinates into RES by the existing finite-Gamma balancing relation. Given an RV bijection of product unions, work at the largest remaining free Gamma arity m. On product neighbourhoods of full-rank types, the local generic-fibre calculation gives the form

$$F(x,u) = \big(h(x), c(x)u^M\big), \qquad M \in \mathrm{GL}_m(\mathbb{Z}), \qquad \mathrm{vrv}(c(x)) = \boldsymbol{\delta},$$

where h is a RES bijection and the value tuple $\boldsymbol{\delta}$ is constant and definable over $\mathbb{S}$. The coefficients $c(x)$ are retained as functions of x. The image is exactly $h(X_0) \times (MU_0 + \boldsymbol{\delta})^\sharp$. The unimodular part is already allowed in $\Gamma[*]$; its translation part is removed one coordinate at a time by the displayed generators. Compactness and finite product refinement match finitely many disjoint such rectangles. The remainder has lower Gamma rank on both sides, so induction applies. At rank zero, every Gamma factor has been absorbed into RES and the equality is a RES isomorphism. This argument uses no cancellation and makes no same-ground choice of the coefficients.

For the positive semiring, replace "ideal generated by the differences" with "congruence generated by the pairs"

$$[X]_r \otimes [U]_m \;\sim\; [X]_r \otimes [U + \delta e_i]_m.$$

Groupification then gives the stated ideal, without assuming that the positive monoid is cancellative. In particular, an arbitrary kernel element can first be written $a - b$; equality after groupification is a positive equality after adding a common RV class, which can be lifted to a positive tensor by surjectivity.

A single uniform annihilator. In the graded tensor put

$$\tau = 1 \otimes [\{0\}]_1 = [\mathbb{G}_m]_1 \otimes 1. \qquad \text{Then} \qquad \tau \ker\Psi = 0.$$

Indeed c gives $X \times 0^\sharp \simeq X \times \delta^\sharp$. Balance the latter degree-one coefficient to the Gamma side and use the integral shear $(\alpha, \gamma) \mapsto (\alpha, \gamma + \alpha e_i)$ on the slice $\alpha = \delta$. Balance back to obtain $\tau g_{X,c,U,i} = 0$. Thus $\Psi[\tau^{-1}]$ is an isomorphism. Here τ has degree one and cannot be cancelled in the original ring. The stronger equality $\ker\Psi = \mathrm{Ann}(\tau)$ is not asserted.

The previously found nonzero element is the special generator

$$[\{r : r^5 = \mathrm{rv}(t)\}]_0 \otimes \big([(0,\infty)]_1 - [(\tfrac{1}{5},\infty)]_1\big), \qquad c(r) = r,$$

over $\mathbb{S} = \mathbb{C}((t))$, $\mathrm{val}(t) = 1$. The detailed mathematical proof, including the local rectangle and lower-rank arguments, is recorded in the section's verification files. The complete uniform compactness and scissor argument remains to be exported in Lean. No HK theorem is taken as a premise, no statement about the measured kernel is inferred, and the working manuscript has not been changed.

Note that, however, [8, Proposition 10.10(2)] does not hold. One of the main motivations of this paper is to construct an alternative to be used in applications.

Notation 3.19. For simplicity, we often drop the constant volume form 0 from the notation. For instance, if $\boldsymbol{U}$ is an object of $\mathrm{RV}[*]$ then it may also denote the object $(\boldsymbol{U}, 0)$ of $\mu\mathrm{RV}[*]$ with the constant volume form 0.

Notation 3.20. Let $[1] \in \mathbf{K}_+\,\mathrm{RES}[1]$ be the class of the singleton $\{1\}$. The class of the singleton $\{1\}$ in $\mathbf{K}_+\,\mathrm{RES}[0]$ is the multiplicative identity of $\mathbf{K}_+\,\mathrm{RES}[*]$ and hence is simply denoted by 1. We have $[\mathrm{RV}^{\circ\circ}_\infty] = [\mathrm{RV}^{\circ\circ}] + 1$ in $\mathbf{K}_+\,\mathrm{RV}[\leq 1]$.

Let I_{sp} be the (nonhomogeneous) semiring congruence relation on $\mathbf{K}_+\,\mathrm{RV}[*]$ generated by the pair $([1], [\mathrm{RV}^{\circ\circ}_\infty])$. Let

$$\boldsymbol{P} = [1] - [\mathrm{RV}^{\circ\circ}] \in \mathbf{K}\,\mathrm{RV}[1].$$

The corresponding principal ideal of $\mathbf{K}\,\mathrm{RV}[*]$ is thus generated by the element $\boldsymbol{P} - 1$.

Similarly, let $\mu\mathrm{I}_{\mathrm{sp}}$ be the semiring congruence relation on $\mathbf{K}_+\,\mu\mathrm{RV}[*]$ generated by the pair $([1], [\mathrm{RV}^{\circ\circ}])$, which is homogeneous, and the corresponding principal ideal of $\mathbf{K}\,\mu\mathrm{RV}[*]$ is generated by the element $\boldsymbol{P}$.

Resolved referee R051 [Typo] – Spelling of homogeneous.
Original source line 1358. In this notation, replace "nonhomogenous" and "homogenous" by "nonhomogeneous" and "homogeneous," respectively.

Author resolution. The author approves correcting the spellings to nonhomogeneous and homogeneous. The congruences and their generators are unchanged.

Notation 3.21. For each $\boldsymbol{U} = (U, f) \in \mathrm{RV}[k]$, let U_f be the set $\bigcup\{f(u)^\sharp \times u \mid u \in U\}$. Let $\mathbb{L}_{\leq k} : \mathrm{RV}[\leq k] \longrightarrow \mathrm{VF}[k]$ be the map given by $\boldsymbol{U} \longmapsto U_f$. Set $\mathbb{L} = \bigcup_k \mathbb{L}_{\leq k}$.

Let $\mu\mathbb{L}_k : \mu\mathrm{RV}[k] \longrightarrow \mu\mathrm{VF}[k]$ be the map given by $(\boldsymbol{U}, \omega) \longmapsto (\mathbb{L}\boldsymbol{U}, \mathbb{L}\omega)$, where $\mathbb{L}\omega$ is the obvious function on $\mathbb{L}\boldsymbol{U}$ induced by ω. Set $\mu\mathbb{L} = \bigoplus_k \mu\mathbb{L}_k$.

Suppose that $\mathbb{S}$ is (VF, Γ)-generated. The map $\mathbb{L}$ induces a surjective semiring homomorphism $\mathbf{K}_+\,\mathrm{RV}[*] \longrightarrow \mathbf{K}_+\,\mathrm{VF}_*$ and the map $\mu\mathbb{L}$ induces a surjective graded semiring homomorphism $\mathbf{K}_+\,\mu\mathrm{RV}[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{VF}[*]$, see [8, § 4, § 6] or [17, Corollaries 7.7, 10.6]; we use the same notation for these homomorphisms as well as their groupifications. We have

Resolved referee R053 [Error; revised] – Scope of the parameter hypothesis.
Original source line 1366. State here the (VF, Γ)-generation hypothesis used in Theorem 3.22, before asserting that lifting induces these homomorphisms. Without a parameter restriction, even well-definedness can fail. For example, let the parameters be generated by a trivially valued algebraically closed field C and a named $\tau \in \mathrm{RV}$ of positive value. The singleton objects $\{\tau\}$ and $\{1\}$ are isomorphic in $\mathrm{RV}[1]$. But $\tau^\sharp$ has no definable VF point: Lemma 2.13 places every such point in $\mathrm{acl}(C) \cap \mathrm{VF} = C$. In contrast, $1 + \mathcal{M}$ contains the definable point 1, so their lifts cannot be definably bijective. Moving the stated hypothesis forward removes this obstruction.

Author resolution. The author approves stating the (VF, Γ)-generation hypothesis before both homomorphism assertions. The lift on individual objects needs no such hypothesis. Under the standing nontrivial-ground-value-group assumption, take instead a small algebraically closed nontrivially valued subfield F and name $\tau \in \mathrm{RV}$ with $\mathrm{vrv}(\tau) \notin \Gamma(F)$. Lemma 2.13 puts every definable VF point over the